INCH -POUND

MIL-DTL-32291 (OS)
01 July 2008

DETAIL SPECIFICATION

CARTRIDGE, IMPULSE, CCU-165/B

*This specification is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.*

## 1. SCOPE

**1.1 Scope.** This specification covers the requirements for the manufacture and acceptance of the Cartridge, Impulse, CCU-165/B, referred to in this document as cartridge (see 6.1).

**1.2 Application.** The cartridges covered by this specification are used in various racks and launchers to provide gas pressure to activate the gas actuated pistons in the racks and launchers to eject various stores from aircraft.

## 2. APPLICABLE DOCUMENTS

**2.1 General.** The documents listed in this section are specified in sections 3, 4, or 5 of this specification. This section does not include documents cited in other sections of this specification, or documents recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements for documents referenced in sections 3, 4, or 5 of this specification, whether or not they are listed.

**2.2 Government documents**.

**2.2.1 Specifications, Standards, and Handbooks.** This section is not applicable to this document

**2.2.2 Other Government documents, drawings and publications.** The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation or contract.

Comments, suggestions, or questions on this document should be addressed to DEPARTMENT OF THE NAVY, Indian Head Division, NSWC, Code E11G3, Document Control, 4072 North Jackson Road Suite 106, Indian Head, MD 20640-5115 OFFICIAL BUSINESS, or emailed to amanda.penn@navy.mil. Since contact information can change, you may want to verify the currency of this information using the ASSIST Online database at http://assist.daps.dla.mil.

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DRAWINGS

NAVAL AIR SYSTEMS COMMAND (30003)

| | |
|---|---|
| 3205AS357 | Cartridge, Impulse Assembly |
| 3205AS325 | Cartridge Case and Ignition Element Assembly (Modified) |
| 3205AS359 | Header, Glass Sealed |
| 3205AS319 | Header and Bridge Assembly |
| 3205AS318 | Ignition Element Loaded Assembly |
| 3205AS358 | Header, Bridgewire and Charge Holder Assembly |
| 3205AS320 | Header, Glass Seal and Capacitor Assembly |
| DL 838AS123 | Closed Bomb Assembly |
| 834AS207 | Smokeless Powder |
| 838AS206 | Propellant Charge, Double Base |

(Application for copies should be addressed to DEPARTMENT OF THE NAVY, Indian Head Division, NSWC, Code E11G7, Document Control, 4072 North Jackson Road Suite 106, Indian Head, MD 20640-5115 OFFICIAL BUSINESS.)

**2.3 Non-Government publications and drawings**. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are the issues of the documents cited in the solicitation or contract.

AMERICAN SOCIETY FOR TESTING MATERIALS

| | |
|---|---|
| ASTM E 1742 | Standard Practice for Radiographic Inspection (DoD adopted) |

(Copies of these documents are available online at http://www.astm.org/ or from the American Society for Testing and Materials Customer Service, 100 Barr Harbor Drive, West Conshohocken, PA 19428-2959.)

ASSOCIATION CONNECTING ELECTRONICS INDUSTRIES

| | |
|---|---|
| IPC-J-STD-001 | Requirements for Soldered Electrical and Electronic Assemblies |

(Copies of these documents are available online at http://www.ipc.org/ or from the



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IPC-Association Connecting Electronics Industries, 3000 Lakeside Drive, 309 S, Bannockburn, IL 60015, PH 847-615-7100, FAX 847-615-7105.)

**2.4 Order of Precedence**. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

## 3. REQUIREMENTS

**3.1 Cartridges.** Cartridges shall be manufactured to the requirements of this specification and drawing 3205AS357.

**3.2 Ignition elements.** Ignition elements submitted for first article testing shall meet the requirements of drawing 3205AS318.

**3.3 First article.** When specified in the contract or purchase order (see 6.2), a sample shall be subjected to first article inspection (see 4.2.1 and 6.3).

**3.4 Cartridge production and primary components.** The cartridge shall be manufactured in a continuous process. For the purposes of this specification, the header glass seal, drawing 3205AS359, the ignition element, drawing 3205AS318, and the main propellant charges, drawings 834AS207 and 838AS206 are defined as primary components. Only one lot of a primary component shall be used in the manufacture of a single lot of cartridges.

**3.5 Header pressure integrity.** The Header, Glass Sealed, 3205AS359, shall be capable of withstanding 30,000 ± 500 psi hydrostatic pressure applied to the bridge wire side of the assembly without any leakage, cracking of the glass or movement or deformation of any parts.

**3.6 Header dead load.** The Header, Bridgewire and Charge Holder Assembly, 3205AS358 shall be capable of withstanding the 300 pound force dead load applied to the electrode without cracking the glass and without movement or deformation of the parts.

**3.7 Header insulation resistance.** The Header, Glass Sealed, 3205AS359, shall have an insulation resistance greater than 50 megohms.

**3.8 Soldering.** Soldering shall meet the requirements of IPC-J-STD-001

**3.9 Visual.** Cartridges and ignition elements shall be free of visual defects such as inadequate or illegible markings, burrs, dents, deep scratches, split or cracked crimp, wrinkled or creased mylar closure or closure disks or other defects which may prevent proper assembly or environmental stability of the parts.

**3.10 Radiographic.** Cartridges and ignition elements shall be free of imperfections in components or assembly as verified by radiographic examination.



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**3.11 Resistance.** The resistance of the bridge circuit of each cartridge, header and bridge wire assembly and ignition element loaded assembly shall be as stated on their respective drawings.

**3.12 Impedance.** The bridge filter for each cartridge or ignition element shall have a peak to peak voltage of 40 millivolts maximum when tested as described in 4.3.4.

**3.13 Power–current test.** Ignition elements and cartridges shall be capable of dissipating the heat generated by 1 amp of current without firing or degrading, and shall meet the functional test requirement of section 3.18 or 3.19 respectively.

**3.14 Leakage.** Ignition elements when tested in a dry gas leak tester shall have a leak rate of less than $10^{-4}$ at .5 atm pressure differential.

**3.15 Six-foot drop .** Cartridges shall be capable of being dropped unprotected from a height of six feet onto a two inch minimum thickness steel plate embedded in concrete without impairing their ability to perform their intended function.

**3.16 Twenty G shock.** Cartridges shall be capable of withstanding a 20G peak saw tooth shock of 11 msec duration and meet the functional requirement of section 3.18.

**3.17 Temperature and humidity cycling.** Cartridges shall withstand temperature and humidity cycling between -80, -65, 70°F at ambient humidity and 160°F at 90% humidity and meet the functional requirement of section 3.19.

**3.18 Vibration.** Cartridges shall withstand the vibration test of paragraph 4.3.15 and meet the functional requirement of section 3.19.

**3.19 Ignition element ballistic performance.** The ignition element, drawing 3205AS318 when tested as part of the cartridge case and ignition element assembly (modified), drawing 3205AS325, shall meet the requirements of paragraphs 3.19.1 and 3.19.2 when fired in the closed bomb DL 838AS123.

**3.19.1 Ignition delay.** The maximum allowable ignition delay is ten milliseconds (see 6.6.1).

**3.19.2 Maximum pressure.** The minimum allowable value for maximum pressure (see 6.6.5) shall be 2,000 pounds per square inch (psi).

**3.20 Cartridge Ballistic performance.** Cartridges shall propel a 480 ±5 pound mass horizontally when fired in the variable volume test fixture at an initial volume of 3.7 cubic inches while maintaining the following parameters.

**3.20.1 Ignition delay.** Ignition delay time shall not exceed ten milliseconds (see 6.6.1).

**3.20.2 Time to maximum thrust.** The time to maximum thrust shall not exceed 30 milliseconds (see 6.6.2).



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**3.20.3 Thrust upper limit.** The upper limit of thrust shall be 12,200 pounds force (lbf) (see 6.6.4)

**3.20.4 Velocity lower limit.** The lower limit for velocity shall not be less than 22 feet per second (see 6.6.3).

**3.21 Workmanship.** Cartridges shall be constructed and finished in a manner to assure compliance with all requirements of this specification. Particular attention should be directed to dimensions, finishes, crimping and assembly operations.

## 4. VERIFICATION

**4.1 Classification of inspections**. The inspection requirements specified herein are classified as follows:

a. First Article inspection (see 4.2.1).

b. Lot production and in-process tests (see 4.2.2)

c. Acceptance tests (see 4.2.3)

**4.2 Inspections.**

**4.2.1 First article inspection.** First article inspection shall be required under the following conditions.

a. First time vendor,
b. Two or more years has passed since the CCU-165/B or similar items have been manufactured,
c. There has been a major turn over in personnel,
d. There have been major changes to the production facility or
e. There has been a transfer of production to a new facility or site.

First article inspection sample shall be comprised of 70 cartridges, drawing 3205AS357; 25 header, glass sealed assemblies, drawing 3205AS359; 25 header, bridgewire and charge holder assemblies, drawing 3205AS358 and 25 cartridge case and ignition element assembly (modified), drawing 3205AS325. Units shall be prepared using the same procedures and methods proposed for manufacture of the production lot and subjected to the tests listed in Table I. Tests shall be conducted in the top to bottom order of Table I. All units must pass these tests, no failures are allowed.

**4.2.2 Lot production and in-process tests.**

**4.2.2.1 Lot size and acceptance test sample size**. Unless otherwise specified cartridge production lot size is limited to 20,000 maximum excluding test cartridges. Acceptance test sample size will be 53 with 50 expended in testing and 3 retained for investigation purposes.



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**4.2.2.2 In-process tests and inspections.** The manufacturer shall conduct the in-process inspections on the levels of assembly presented in Table II on the complete production lot (100% inspection). Reject any units that fail these tests.

TABLE I. First article tests and conditions.

| Item | Test | Para. | Cartridges | | | | | | Header w/ Charge Holder | Header Glass Seal | Cart Case Ignition Element (modified) |
|---|---|---|---|---|---|---|---|---|---|---|---|
| Cart/Header/Ign element | Visual | 4.3.1 | 6 | 12 | 18 | 12 | 18 | 4 | 25 | 25 | 25 |
| Cart/Ign element | Resistance | 4.3.3 | 6 | 12 | 18 | 12 | 18 | 4 | 25 | | 25 |
| Cart/Ign element | Impedance | 4.3.4 | 6 | 12 | 18 | 12 | 18 | 4 | 25 | | 25 |
| Cart/Ign element | Radiographic | 4.3.2 | 6 | 12 | 18 | 12 | 18 | 4 | | | 25 |
| Cartridge | Six-foot drop | 4.3.12 | 6 | | | | | | | | |
| Cartridge | Twenty G shock | 4.3.13 | | 12 | | | | | | | |
| Cartridge | T & H Cycling | 4.3.14 | | | 18 | | | | | | |
| Cartridge | Vibration | 4.3.15 | | | | 12 | | | | | |
| Cartridge | Gauge | 4.3.6 | 6 | 12 | 18 | 12 | | | | | |
| Cartridge | Electrode Recess | 4.3.7 | 6 | 12 | 18 | 12 | | | | | |
| Cartridge | Resistance | 4.3.3 | 6 | 12 | 18 | 12 | | | | | |
| Cartridge | Impedance | 4.3.4 | 6 | 12 | 18 | 12 | | | | | |
| Cartridge | Radiographic | 4.3.2 | 6 | 12 | 18 | 12 | | | | | |
| Cartridge | Test fire 70°F | 4.3.16.2 | 6 | 12 | 18 | 12 | | | | | |
| Cartridge | Test fire -65°F | 4.3.16.2 | | | | | 9 | | | | |
| Cartridge | Test fire 200°F | 4.3.16.2 | | | | | 9 | | | | |
| Cartridge | Retain sample | | | | | | | 4 | | | |
| Header Bridge Wire and Charge Holder Assy | Dead load | 4.3.10 | | | | | | | 25 | | |
| Cart/Ign element | Resistance | 4.3.3 | | | | | | | 25 | | |
| Cart/Ign element | Impedance | 4.3.4 | | | | | | | 25 | | |
| Header Glass Seal | Header pressure | 4.3.9 | | | | | | | | 25 | |
| Header Glass Seal | IR | 4.3.11 | | | | | | | | 25 | |
| Ignition element | Leakage | 4.3.5 | | | | | | | | | 25 |
| Ignition element | Power Current | 4.3.8 | | | | | | | | | 25 |
| Ignition element | Test fire 70°F | 4.3.16.1 | | | | | | | | | 25 |

Table II In-process Inspections

| Test | Paragraph | level of assembly | % inspection |
|---|---|---|---|
| Resistance | 4.3.3 | Header and Bridgewire (3205AS319)<br>Ign element loaded (3205AS318)<br>Cartridge (3205AS357) | 100 |
| Impedance | 4.3.4 | Header w/capacitor (3205AS320)<br>Ign element loaded (3205AS318)<br>Cartridge (3205AS357) | 100 |
| Leakage | 4.3.5 | Ign element loaded (3205AS318) | 100 |
| Visual | 4.3.1 | Cartridge (3205AS357) | 100 |
| Gauge | 4.3.6 | Cartridge (3205AS357) | 100 |



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**4.2.3 Acceptance tests.** Acceptance tests and inspections are presented in Table III. Tests shall be conducted in the top to bottom order of Table III. All units must pass these tests, no failures are allowed.

**4.2.4 Misfire.** A cartridge that fails to ignite during testing shall be considered unacceptable unless the failure to ignite is clearly attributable to faulty test equipment or procedure.

**4.2.5 Test failure.** If a test failure is attributed to faulty test equipment or test setup the original test results shall be discarded and that test reconducted, otherwise no failures are allowed.

TABLE III. Acceptance tests and examinations.

| Item | Test | paragraph | qty |
|---|---|---|---|
| Cartridge | Visual | 4.3.1 | 50 |
| Cartridge | Diameter Gauge | 4.3.6 | 50 |
| Cartridge | Electrode Recess | 4.3.7 | 50 |
| Cartridge | Radiograph | 4.3.2 | 50 |
| Cartridge | Resistance | 4.3.3 | 50 |
| Cartridge | Impedance | 4.3.4 | 50 |
| Cartridge | Ballistic | 4.3.16.2 -65°F | 25 |
| | | 4.3.16.2 200°F | 25 |

**4.2.6 Change in propellant lot(s).** Whenever a change occurs in the propellant lot, test firings shall be conducted to ensure cartridges are within the limits specified in 3.19.

**4.3 Test Requirements**

**4.3.1 Visual examination.** Cartridges and ignition elements shall not have inadequate or illegible markings or serial numbers, dents, deep scratches, split or cracked edges, chipped surfaces, or other defects which may prevent proper installation or function.

**4.3.2 Radiographic inspection.** Cartridges and ignition elements shall be x-rayed, in accordance with ASTM E 1742, to verify the presence, condition and proper installation of parts. Any observable imperfections shall be cause for rejection. All units shall be identified with serial numbers prior to radiographic inspection. Samples shall be arrayed on trays or boards in consecutive numerical order and shall be positioned to give the most revealing cross-sectional image. Each radiograph shall carry the permanent identification of the items displayed. The radiographic identification shall include the complete lot number and the span of serial numbers displayed.

**4.3.3 Resistance.** Conduct tests on the levels of assembly identified in Tables I and II using a test current of less than 25 mA. Resistance values shall be as stated on the respective drawings.

**4.3.4 Impedance**. The signal generator shall be calibrated with a 1-ohm resistor load to show an alternating current 100.0 ± 0.01 millivolts peak to peak, at a frequency of 1.0 ± 0.01 megahertz. Remove the resistor and replace with the cartridge or ignition element as applicable. Apply the



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current for 10 milliseconds maximum between the cartridge/ignition element body and the contact pin through a 47 ± 10% Ohm, 0.25 Watt minimum resistor. The voltage drop shall be monitored using an oscilloscope set to the appropriate scale. The oscilloscope shall be capable of a minimum of 20 meg-samples per second and shall have a 50-Ohm terminator.

With the ignition element assembly installed in place of the calibrated 1 Ohm resistor, the voltage shall be 40 millivolts peak to peak maximum.

**4.3.5 Leakage.** Ignition elements shall be leak tested in a dry gas leak tester to determine conformance with 3.14.

**4.3.6 Gauge test.** Pass each cartridge through a max diameter size gauge to a depth of 1 inch to verify cartridges do not exceed the maximum allowable diameter at the roll crimp as shown on drawing 3205AS357.

**4.3.7 Electrode Recess.** Using a depth gauge measure the recess of the electrode, surface “B”, as shown on drawing 3205AS357.

**4.3.8 Power–current test.** Apply 1 amp of current to the bridge circuit and maintain for 5 minutes. The test current shall be regulated throughout the period of application to within two percent. The test temperature is ambient. Units shall not fire or degrade and shall meet all subsequent test requirements.

**4.3.9 Header pressure integrity test.** Apply 30,000 ± 500 psi of hydrostatic pressure to surface “C” of the glass sealed header as shown on drawing 3205AS359. Headers shall meet the requirements of paragraph 3.5.

**4.3.10 Header dead load.** Apply 300 pound force minimum to surface “B” of the electrode on the Header, Bridgewire and Charge Holder Assembly, drawing 3205AS358. Headers shall meet the requirements of paragraph 3.6

**4.3.11 Insulation resistance.** Measure the insulation resistance between the electrode at surface “B” and the header as shown on the glass sealed header drawing 3205AS359. Apply 500 volts D.C. between the electrode surface “B” and the header. Units shall meet the requirement of paragraph 3.7

**4.3.12 Six-foot drop test**. A sample of 6 cartridges shall be selected at random from the first article lot. Two cartridges shall be dropped in each of the following positions; (1) Ignition Element side up, (2) Ignition Element side down, and (3) Ignition Element horizontal. A new cartridge shall be used for each drop and no cartridge shall fire during this test. Cartridges shall be free from damage and shall meet all ballistic requirements of section 3.19.

**4.3.13 Twenty G shock.** A sample of 12 cartridges shall be selected at random from the first article lot. Cartridges shall be mounted in a test fixture and subjected to 3 shocks in each direction of 3 mutually perpendicular axes for a total of 18 shocks. The wave form shall be as



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given in 3.15. Cartridges shall be free from damage and shall meet all ballistic requirements of section 3.19.

**4.3.14 Temperature and humidity cycling.** This test provides for cycling between temperatures of -80°F, –65°F, +70°F at ambient humidity and 160°F at 90 percent relative humidity, with additional storage at –80°F and +160°F for two periods of two weeks each. Provision has been made for withdrawal of cartridges at two times during the cycling, see schedule Table IV. This is to provide opportunity to observe the extent of progressive deterioration, if such exits. Cartridges shall be supported on screen trays so that all areas are exposed to the prescribed atmospheric conditions at all times throughout the test. At the conclusion of the test, cartridges must be operable, free from damage and shall meet all ballistic requirements of section 3.19. The schedule to be followed is provided in Table IV.

**4.3.15 Vibration**. Vibration testing shall be comprised of frequency cycling and resonance cycling. Testing shall be conducted at three temperatures, ambient, -65 and 200°F, and in two orientations, parallel and perpendicular to the axis of the cartridge. Mount the units in a vibration fixture and subject them to frequency cycling between 10 and 2000 cps in 15 minute cycles with an amplitude of 0.018 inch (total excursion 0.036 inch) or an applied acceleration of ± 10 G, whichever is the limiting value at the temperatures and durations given in Table V for each orientation. Resonance cycle the units at 55 cps and applied amplitude of 0.030 inch (total excursion of .060 inch) at the temperatures and durations given in Table V. Any order of orientation, temperature and test, may used as long as the full duration for each orientation, temperature and test is achieved. Cartridges shall be free from damage and shall meet all ballistic requirements of section 3.19.

**4.3.16 Ballistic testing.**

**4.3.16.1 Ignition element ballistic test.** Conduct ignition element ballistic testing at the temperature listed in Table I. Temperature condition units for 4 hours. Fire the sample in the Closed Bomb Assembly, DL 838AS123, within 3 minutes of removal from conditioning using a 4.0 ± 0.1 amp current. Ignition elements shall meet the requirements of paragraph 3.18. No failures are allowed.

**4.3.16.2 Cartridge ballistic test.** Cartridges shall be temperature conditioned at the temperatures shown in Tables I or III as applicable for 4 hours minimum. Fire the cartridges in the variable volume test fixture within 3 minutes of removal from the chamber. Cartridges shall meet the requirements of paragraph 3.19, no failures are allowed. In addition breach pressure shall be recorded, the expected pressure range is 7300 to 9800 psi. Pressure is recorded for informational purposes only.



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Table IV. Temperature and humidity schedule.

| DAY | TIME | CONDITION |
|---|---|---|
| Monday | 1300 | Place in cabinet or room maintained at –65°F; |
| | 1600 | Remove from –65°F room and place in room maintained at +160°F, and 90 percent relative humidity and allow to remain overnight; |
| Tuesday | 0800 | Remove from +160°F and place in +70°F; |
| | 1300 | Remove from +70°F and place in –65°F; |
| | 1600 | Remove from –65°F and place in +160°F and 90 percent relative humidity |
| Wednesday | 0800 | Remove from +160°F and place in +70°F; |
| | 1300 | Remove from +70°F and place in –65°F; |
| | 1600 | Remove from –65°F and place in +160°F and 90 percent relative humidity |
| Thursday | 0800 | Remove from +160°F and place in +70°F; |
| | 1300 | Remove from +70°F and place in –65°F; |
| | 1600 | Remove from –65°F and place in +160°F and 90 percent relative humidity |
| Friday | 0800 | Remove from +160°F and place in +70°F; |
| | 1300 | Remove from +70°F and place in –65°F; |
| | 1600 | Remove from –65°F and place in +160°F and 90 percent relative humidity |
| Saturday and Sunday | | Maintain in +160°F and 90 percent relative humidity |
| Monday | 0800 | Remove from +160°F and place in +70°F; |
| | 1300 | Remove from +70°F and place in –65°F; |
| | 1600 | Remove from –65°F and place in +160°F and 90 percent relative humidity |
| Tuesday | 0800 | Remove from +160°F and place in +70°F; |
| | 1300 | Remove from +70°F and place in –65°F; |
| | 1600 | Remove from –65°F and place in +160°F and 90 percent relative humidity |
| Wednesday | 0800 | Remove from +160°F and place in +70°F; |
| | 1300 | Remove from +70°F and place in –65°F; |
| | 1600 | Remove from –65°F and place in +160°F and 90 percent relative humidity |
| Thursday | 0800 | Remove from +160°F and place in +70°F; |
| | 1300 | Remove from +70°F and place in –65°F; |
| | 1600 | Remove from –65°F and place in +160°F and 90 percent relative humidity); |
| Friday | 0800 | Remove from +160°F and place in +70°F; |
| | 1300 | Remove from +70°F and place in –65°F; |
| Saturday and Sunday | | Maintain at -80°F |
| Monday | 0800 | Remove from -80°F and place in +70°F. Remove six cartridges and store at +70°F which are to be fired at the end of cycling program. |
| The second two-week period follows the same schedule of temperature and humidity cycling. Withdrawals of cartridges during second two-week period are to be as follows: | | |
| Fourth Monday | 0800 | Remove from +160°F and place in +70°F. Remove six cartridges and store at +70°F, which shall be fired at the end of cycling program. |
| The completion of the two periods occurs at 0800 on the fifth Monday after starting. At this time the remaining six cartridges are to be removed, conditioned at +70°F, together with remainder of cartridges from two and three weeks cycling, and fired. | | |



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TABLE V. Vibration test schedule.

| Type | Vibration at Room Temperature | Vibration at +200° F | Vibration at –65° F |
|---|---|---|---|
| | ***(Times at each position: up, horizontal)*** | | |
| Cycling | 60 minutes | 15 minutes | 15 minutes |
| Resonance | 120 minutes | 30 minutes | 30 minutes |

## 5. PACKAGING

**5.1 Packaging.** For acquisition purposes, the packaging requirements shall be as specified in the contract or order, (see 6.2). When actual packaging of material is to be performed by DOD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military Service or Defense Agency, or within the military service' system command. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

## 6. NOTES

(This section contains information of a general or explanatory nature that may be helpful but is not mandatory.)

**6.1 Intended use**. CCU-165/B Cartridges, Impulse, covered by this specification are used in various racks and launchers to provide gas pressure to activate the gas actuated pistons in the racks and launchers to eject various stores from military aircraft. These impulse cartridges are military unique and there is no equivalent commercial application for this design.

**6.2 Acquisition requirements.** Acquisition documents should specify the following:

a. Title, number and date of this specification.

b. Issue of DODISS to be cited in the solicitation and, if required, the specific issue of individual documents referenced.

c. Inspection conditions, if other than as specified. (see 4.2)

d. Whether first article inspection is required and, if so, specify the test activity (see 3.3 and 4.2.1).

e. Production lot size (see 4.2.2.1) and test activity.

f. Safety requirements. (see 6.4).

g. Packing requirements (see 5.1).



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h. Special marking requirements, if different from 6.5.2.1 (see 5.1).

**6.3 First article.** When first article inspection is required, the contracting officer should provide specific guidance to offerors whether the item(s) should be a preproduction sample, a first article sample, a first production item, a sample selected from the first production items, or a standard production item from the contractor's current inventory, and the number of items to be tested (see 4.2.1). The contracting officer should also include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results, and disposition of first articles.

**6.3.1 Bids.** Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

**6.4 Ammunition and explosive safety.**

**6.4.1 Contractor's responsibility.** The safety requirements of the "Contractor's Safety Manual for Ammunition, Explosives and Related Dangerous Material" (DOD 4145.26M) are applicable when this specification is used for acquisition.

**6.5 Contract packaging, packing and marking.** The following requirements should be specified in the contract.

**6.5.1 Packaging and packing.** Packaging and packing should be in accordance with 49 CFR 100-199.

**6.5.2 Marking**

**6.5.2.1 Special marking.** Marking of outer containers should be in accordance with CFR 49 171-199.

**6.5.2.2 Normal marking.** In addition to any special marking required by the contract or order, unit packages, intermediate packages and shipping containers should be marked in accordance with MIL-STD-129.

**Inner container marking.**

NSN: DODIC: JM92

Nomenclature: Cartridge, Impulse, CCU-165/B

Part Number: 30003-3205AS357



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Lot Number: (In accordance with MIL-STD-1168)

Hazard Classification 1.4C

**Outer container marking.**

NSN: DODIC: JM92

Nomenclature: Cartridge, Impulse, CCU-165/B

Part Number: 30003-3205AS357

Quantity:

Gross weight and cube:

Lot number: (In accordance with MIL-STD-1168)

Contract or purchase order number:

The above information should be applied in accordance with the provisions of MIL-STD-129.

**6.6 Definitions.**

**6.6.1 Ignition delay.** Ignition delay is the time from application of the current to first indication of continuous thrust on the thrust versus time curve.

**6.6.2 Time to maximum thrust.** Time to maximum thrust is the time from application of current to the point of maximum thrust on the thrust versus time curve.

**6.6.3 Velocity.** Velocity is defined as the time taken for the weight to travel one foot after separation from the barrel. The velocity is in feet per second.

**6.6.4 Thrust.** The value for thrust is the maximum value of pounds force registered on the thrust cell during sled movement.

**6.6.5 Maximum pressure.** Maximum pressure is that pressure recorded by the strain gauge during piston stroke.

**6.7 Subject term (key word) listing.**

Aircraft Ejection seat
Emergency egress
Gas Generator



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Custodian:
Navy - OS

Preparing Activity:
Navy - OS
(Project No. 1377-2008-002)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at http://assist.daps.dla.mil/.



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